KCE PATHWAYS · GRADE 3 · UNIT 4: PARTS AND PIECES · WEEK 18 · MATH

Quilting Day in Gee's Bend — Teacher Lesson Plan & Answer Key

Learning objectives: Students compare two fractions with the same denominator by comparing numerators, compare two fractions with the same numerator by reasoning about piece size, record comparisons with <, > and =, explain why comparisons only make sense when the wholes are the same size, and solve two-step fraction word problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce quilting day in Gee's Bend, Alabama, with Aunt Loretta, Tasha, Calvin and Ruthie.
2. Warm-Up, Part 1: Number Lines & Equivalent Fractions (8 pts, Week 17 skill): a point on a number line and an equivalent-fraction choice.
3. Warm-Up, Part 2 (4 pts): typing the missing numerator in an equivalent pair (for example, 2/3 = ___/6).
4. Picture It: Quilting Day in Gee's Bend. Every strip is the same size, so pieces can be compared fairly. Ask: which is bigger, one of 3 equal pieces or one of 6?
5. Word Wrangler: 6 vocab cards (compare, numerator, denominator, unit fraction, greater than (>), less than (<)). Tip: the open side of the symbol faces the bigger number.
6. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W18.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
7. Same Denominator, Part 1: Compare the Numerators (8 pts): typed <, >, = items and ordering same-denominator strips. Tip: same-size pieces, so more pieces means more fabric.
8. Same Denominator, Part 2 (4 pts): a reasoning item.
9. Same Numerator, Part 1: Compare the Piece Sizes (8 pts): typed comparisons like 1/4 and 1/8 and a which-explanation item. Tip: more equal parts in the whole means smaller pieces.
10. Same Numerator, Part 2 (4 pts): one more same-numerator comparison.
11. Using <, >, and = With Reasons, Part 1 (8 pts): equal fractions such as 4/4 = 8/8 and a reasoning item about wholes of different sizes (half of a bed quilt is not half of a doll quilt).
12. Using <, >, and = With Reasons, Part 2 (4 pts): one more comparison; check that the wholes are the same size.
13. Who Has More? Part 1: Fraction Word Problems (8 pts): two-step problems (find what is left, then compare) and ordering fractions. Tip: write each fraction first, then compare.
14. Who Has More? Part 2 (4 pts): one more word problem.
15. Bonus Practice, Part 1: The Quilt Border (XP only, not scored).
16. Bonus Practice, Part 2: Squares and Strips (XP only, not scored).
17. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches comparison clues to fractions, Set 2 matches words and symbols to meanings, Set 3 matches comparison rules.

18. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names, places and months, end mark, commas); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
19. Speed Arena, Part 1 (XP only, not scored).
20. Speed Arena, Part 2 and Keyboard Corral (XP only, not scored).
21. Boss Battle, Part 1: The Big Quilt Show (10 pts, 5 each): a two-step comparison of two quilt squares and a which-explanation-is-correct item (for example, why 5/8 > 5/6 is wrong).
22. Boss Battle, Part 2: Hanging the Quilt (10 pts, 5 each): typing a fraction between two others and a find-the-mistake item about equivalent fractions.
23. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W18.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
24. Results (Quilting Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Quilt Hero!” with the congratulations message “Quilt hero! You compared every piece of fabric!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every quilter gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: NUMBER LINES & EQUIVALENT FRACTIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Last week you placed fractions on a number line. This number line from 0 to 1 is split into equal parts. What fraction is at point A?	**3/4** (also accepted: 3 / 4, 3 fourths, three fourths, three-fourths)	There are 4 equal parts between 0 and 1, so each jump is 1/4. Point A is 3 jumps from 0, so it is at 3/4.
w2_s0	Set 1: Which fraction is equivalent to 1/2? (Equivalent means it names the same amount.) [2/4 / 1/4 / 2/3 / 3/8]	**2/4**	2/4 and 1/2 land on the same point on a number line, halfway between 0 and 1. So 1/2 = 2/4.
w3_s0	Set 1: Type the missing numerator to make the fractions equivalent. 2/3 = ___ / 6	**4** (also accepted: 4 sixths, four)	Each third splits into 2 sixths. 2 thirds is 2 × 2 = 4 sixths, so 2/3 = 4/6.
w1_s1	Set 2: This number line from 0 to 1 is split into equal parts. What fraction is at point A?	**5/6** (also accepted: 5 / 6, 5 sixths, five sixths, five-sixths)	There are 6 equal parts between 0 and 1, so each jump is 1/6. Point A is 5 jumps from 0, so it is at 5/6.
w2_s1	Set 2: Find the mistake. A number line from 0 to 1 is split into 3 equal parts. Ruthie labels the first tick mark after 0 as 1/4. What went wrong? [There are 3 equal parts, so the first tick mark is 1/3. / The first tick mark should be 1/2. / Nothing is wrong. It is 1/4. / The first tick mark should be 3/3.]	**There are 3 equal parts, so the first tick mark is 1/3.**	Count the equal parts between 0 and 1, not the tick marks. There are 3 parts, so each jump is 1/3.
w3_s1	Set 2: Type the missing numerator to make the fractions equivalent. 3/4 = ___ / 8	**6** (also accepted: 6 eighths, six)	Each fourth splits into 2 eighths. 3 fourths is 3 × 2 = 6 eighths, so 3/4 = 6/8.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: This number line from 0 to 1 is split into equal parts. What fraction is at point A?	**2/3** (also accepted: 2 / 3, 2 thirds, two thirds, two-thirds)	There are 3 equal parts between 0 and 1, so each jump is 1/3. Point A is 2 jumps from 0, so it is at 2/3.
w2_s2	Set 3: Which explanation is correct? Why are 2/4 and 4/8 equivalent? [They land on the same point on a number line from 0 to 1. / They have the same numerator. / They both have even numbers. / 4/8 is bigger because 8 is bigger than 4.]	**They land on the same point on a number line from 0 to 1.**	Equivalent fractions name the same amount. 2/4 and 4/8 are both at the halfway point between 0 and 1.
w3_s2	Set 3: Type the missing numerator to make the fractions equivalent. 1/2 = ___ / 8	**4** (also accepted: 4 eighths, four)	One half is the same as 4 of 8 equal parts, so 1/2 = 4/8.

SAME DENOMINATOR: COMPARE THE NUMERATORS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1a_s0	Set 1: Type <, >, or = to compare the fractions. 2/6 ___ 5/6	**<** (also accepted: is less than, less than)	Both are sixths, so the pieces are the same size. 2 pieces is fewer than 5 pieces, so 2/6 < 5/6.
p1b_s0	Set 1: Aunt Loretta has three same-size strips. One is 3/8 red, one is 7/8 red, and one is 5/8 red. Type the greatest fraction.	**7/8** (also accepted: 7 / 8, 7 eighths, seven eighths, seven-eighths)	All three are eighths, so compare the numerators: 7 is more than 5 and 3. The greatest is 7/8.
p1c_s0	Set 1: Find the mistake. Ruthie says 3/6 = 5/6 because both fractions have a 6 on the bottom. What went wrong? [The pieces are the same size, but 5 pieces is more than 3 pieces, so 3/6 < 5/6. / Nothing is wrong. They are equal. / 3/6 is greater because 3 comes first. / They are equal because both are less than 1 whole.]	**The pieces are the same size, but 5 pieces is more than 3 pieces, so 3/6 < 5/6.**	The same denominator tells you the pieces are the same size. Then compare the numerators: 3 < 5, so 3/6 < 5/6.
p1a_s1	Set 2: Type <, >, or = to compare the fractions. 7/8 ___ 4/8	**>** (also accepted: is greater than, greater than)	Both are eighths, so the pieces are the same size. 7 pieces is more than 4 pieces, so 7/8 > 4/8.
p1b_s1	Set 2: Type a fraction with a denominator of 4 that is greater than 2/4 but less than 4/4.	**3/4** (also accepted: 3 / 4, 3 fourths, three fourths, three-fourths)	In fourths, the only numerator between 2 and 4 is 3. So the fraction is 3/4.
p1c_s1	Set 2: Which explanation is correct? Why is 4/6 > 1/6? [Both are sixths, so the pieces are the same size, and 4 pieces is more than 1 piece. / Because both have a 6, so the bigger one is the one written first. / Because 4/6 has a bigger denominator. / Because unit fractions are always the smallest fractions.]	**Both are sixths, so the pieces are the same size, and 4 pieces is more than 1 piece.**	Same denominator means same-size pieces. Then the fraction with more pieces, 4/6, is greater.
p1a_s2	Set 3: Type <, >, or = to compare the fractions. 2/3 ___ 1/3	**>** (also accepted: is greater than, greater than)	Both are thirds, so the pieces are the same size. 2 pieces is more than 1 piece, so 2/3 > 1/3.
p1b_s2	Set 3: Type the missing numerator. The fraction must be greater than 4/6 and less than 1 whole. ___ / 6	**5** (also accepted: five)	In sixths, 6/6 is 1 whole. The only numerator more than 4 and less than 6 is 5, so the fraction is 5/6.

ID	Question	Answer	Why (explain to student)
p1c_s2	Set 3: Find the mistake. Calvin put these fractions in order from least to greatest: 5/8, 2/8, 7/8. What went wrong? [2/8 is less than 5/8, so the order should be 2/8, 5/8, 7/8. / Nothing is wrong. The order is correct. / The order should be 7/8, 5/8, 2/8. / Eighths cannot be put in order.]	**2/8 is less than 5/8, so the order should be 2/8, 5/8, 7/8.**	All three are eighths, so order the numerators from least to greatest: 2, 5, 7. That gives 2/8, 5/8, 7/8.

SAME NUMERATOR: COMPARE THE PIECE SIZES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p2a_s0	Set 1: Type <, >, or = to compare the fractions. 1/4 ___ 1/8	**>** (also accepted: is greater than, greater than)	Both are 1 piece. Fourths are bigger pieces than eighths, so 1/4 > 1/8.
p2b_s0	Set 1: Which explanation is correct? Why is 3/8 < 3/4? [Both fractions have 3 pieces, but eighths are smaller pieces than fourths. / Because 8 is greater than 4, so 3/8 is greater. / Because the numerators are the same, so the fractions are equal. / Because 3/8 has more pieces.]	**Both fractions have 3 pieces, but eighths are smaller pieces than fourths.**	Same numerator means the same number of pieces. Eighths are smaller than fourths, so 3/8 < 3/4.
p2c_s0	Set 1: Two steps! Tasha and Ruthie each have a same-size square of fabric. Tasha cuts hers into 6 equal parts and uses 2 parts. Ruthie cuts hers into 3 equal parts and uses 2 parts. Who uses more fabric? Type the name.	**Ruthie** (also accepted: ruthie uses more, ruthie uses more fabric)	Step 1: Tasha uses 2/6 and Ruthie uses 2/3. Step 2: Both are 2 pieces, but thirds are bigger than sixths, so 2/3 > 2/6. Ruthie uses more.
p2a_s1	Set 2: Type <, >, or = to compare the fractions. 2/6 ___ 2/3	**<** (also accepted: is less than, less than)	Both are 2 pieces. Sixths are smaller pieces than thirds, so 2/6 < 2/3.
p2b_s1	Set 2: All of these are parts of same-size wholes: 2/3, 2/8, 2/4. Type the greatest fraction.	**2/3** (also accepted: 2 / 3, 2 thirds, two thirds, two-thirds)	All three have 2 pieces. Thirds are the biggest pieces, because the whole is cut into the fewest parts. The greatest is 2/3.
p2c_s1	Set 2: Find the mistake. Calvin says 1/8 > 1/3 because 8 is bigger than 3. What went wrong? [More parts means smaller parts. An eighth is smaller than a third, so 1/8 < 1/3. / Nothing is wrong. 1/8 is greater. / They are equal because both are unit fractions. / 1/8 is greater because it has more parts shaded.]	**More parts means smaller parts. An eighth is smaller than a third, so 1/8 < 1/3.**	A bigger denominator means the whole is cut into more, smaller pieces. So 1/8 < 1/3.
p2a_s2	Set 3: Type <, >, or = to compare the fractions. 3/4 ___ 3/6	**>** (also accepted: is greater than, greater than)	Both are 3 pieces. Fourths are bigger pieces than sixths, so 3/4 > 3/6.
p2b_s2	Set 3: Ruthie cuts four same-size squares into halves, thirds, fourths, and sixths. Which single piece is the smallest? [1/6 / 1/2 / 1/3 / 1/4]	**1/6**	The square cut into the most equal parts has the smallest pieces. Sixths are the smallest, so 1/6 is the smallest piece.
p2c_s2	Set 3: The denominator is 2, 3, 4, 6, or 8. Type the denominator that makes 2/? greater than 2/6 but less than 2/3.	**4** (also accepted: fourths, 4ths)	With 2 pieces each, a smaller denominator means a greater fraction. The denominator must be between 3 and 6, and the only choice is 4. So 2/6 < 2/4 < 2/3.

USING <, >, AND = WITH REASONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p3a_s0	Set 1: Type <, >, or = to compare the fractions. 4/4 ___ 8/8	**=** (also accepted: is equal to, equal to, equals, equal)	4/4 is 4 of 4 parts, which is 1 whole. 8/8 is also 1 whole. So 4/4 = 8/8.
p3b_s0	Set 1: Which explanation is correct? Calvin says half of the big bed quilt is the same amount of fabric as half of a tiny doll quilt, because 1/2 = 1/2. [He is wrong. The wholes are different sizes, so half of the bed quilt is more fabric. / He is right. 1/2 always means the same amount of fabric. / He is right, because both quilts have 2 halves. / He is wrong, because half of the doll quilt is more fabric.]	**He is wrong. The wholes are different sizes, so half of the bed quilt is more fabric.**	Fractions can only be compared fairly when they are parts of same-size wholes. Half of a bigger quilt is more fabric.
p3c_s0	Set 1: Write this comparison using fractions and a symbol: three fourths is greater than three eighths.	**3/4 > 3/8** (also accepted: 3/4>3/8, 3/8 < 3/4, 3/8<3/4, 3/4> 3/8, 3/4 >3/8, 3/8< 3/4, 3/8 <3/4)	Three fourths is written 3/4, and three eighths is 3/8. Is greater than is >. So 3/4 > 3/8.
p3a_s1	Set 2: Type <, >, or = to compare the fractions. 6/8 ___ 3/4	**=** (also accepted: is equal to, equal to, equals, equal)	Last week you learned that each fourth splits into 2 eighths, so 3 fourths is 6 eighths. 6/8 and 3/4 land on the same point on the number line, so 6/8 = 3/4.
p3b_s1	Set 2: Find the mistake. Ruthie wrote 5/6 < 2/6. What went wrong? [The symbol points the wrong way. 5 sixths is more than 2 sixths, so 5/6 > 2/6. / Nothing is wrong. / She should use =, because both are sixths. / 2/6 is greater because 2 is a smaller number.]	**The symbol points the wrong way. 5 sixths is more than 2 sixths, so 5/6 > 2/6.**	Both are sixths, so compare numerators: 5 > 2. The open side of the symbol must face 5/6, so 5/6 > 2/6.
p3c_s1	Set 2: Write this comparison using fractions and a symbol: one third is less than one half.	**1/3 < 1/2** (also accepted: 1/3<1/2, 1/2 > 1/3, 1/2>1/3, 1/3< 1/2, 1/3 <1/2, 1/2> 1/3, 1/2 >1/3)	One third is 1/3, and one half is 1/2. Is less than is <. So 1/3 < 1/2.
p3a_s2	Set 3: Type <, >, or = to compare the fractions. 3/6 ___ 1/2	**=** (also accepted: is equal to, equal to, equals, equal)	3 of 6 equal parts is half of the whole, the same point on the number line as 1/2. So 3/6 = 1/2.
p3b_s2	Set 3: Which explanation is correct? Tasha's small pillow is 1/4 red. Her big bed quilt is also 1/4 red. Did she use the same amount of red fabric on both? [No. 1/4 of a bigger whole is more fabric than 1/4 of a smaller whole. / Yes, because 1/4 = 1/4. / Yes, because both have 4 equal parts. / No, because the pillow uses more red fabric.]	**No. 1/4 of a bigger whole is more fabric than 1/4 of a smaller whole.**	The fractions match, but the wholes are different sizes. One fourth of the big quilt is more fabric.
p3c_s2	Set 3: Write this comparison using fractions and a symbol: two sixths is less than two thirds.	**2/6 < 2/3** (also accepted: 2/6<2/3, 2/3 > 2/6, 2/3>2/6, 2/6< 2/3, 2/6 <2/3, 2/3> 2/6, 2/3 >2/6)	Two sixths is 2/6, and two thirds is 2/3. Is less than is <. So 2/6 < 2/3.

WHO HAS MORE? FRACTION WORD PROBLEMS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p4a_s0	Set 1: Two steps! Ruthie and Calvin each have a same-size strip cut into 6 equal parts. Ruthie uses 2 parts. Calvin uses 4 parts. Who has more of the strip left? Type the name.	**Ruthie** (also accepted: ruthie has more, ruthie has more left)	Step 1: Ruthie has 6 − 2 = 4 parts left, 4/6. Calvin has 6 − 4 = 2 parts left, 2/6. Step 2: 4/6 > 2/6, so Ruthie has more left.
p4b_s0	Set 1: Put these in order from least to greatest: 3/4, 3/8, 3/6. Type the fraction that goes in the middle.	**3/6** (also accepted: 3 / 6, 3 sixths, three sixths, three-sixths)	All have 3 pieces, so the smallest pieces come first: 3/8, 3/6, 3/4. The middle fraction is 3/6.
p4c_s0	Set 1: Calvin walked 2/3 of the path to the ferry. Ruthie walked 2/4 of the same path. Who walked farther, and why? [Calvin, because both walked 2 parts and thirds of the path are longer than fourths. / Ruthie, because 4 is greater than 3. / They walked the same distance, because both walked 2 parts. / Ruthie, because fourths are longer than thirds.]	**Calvin, because both walked 2 parts and thirds of the path are longer than fourths.**	It is the same path, so the wholes match. Both walked 2 parts, and thirds are longer than fourths. 2/3 > 2/4, so Calvin walked farther.
p4a_s1	Set 2: Two steps! Tasha's quilt row has 8 same-size squares, and she has sewn 3 of them. Calvin's row is the same size, and he has sewn 5 of its 8 squares. Who has more of the row left to sew? Type the name.	**Tasha** (also accepted: tasha has more, tasha has more left)	Step 1: Tasha has 8 − 3 = 5 squares left, 5/8. Calvin has 8 − 5 = 3 squares left, 3/8. Step 2: 5/8 > 3/8, so Tasha has more left to sew.
p4b_s1	Set 2: Put these in order from least to greatest: 2/3, 2/8, 2/4, 2/6. Type the second fraction in your list.	**2/6** (also accepted: 2 / 6, 2 sixths, two sixths, two-sixths)	All have 2 pieces, so the smallest pieces come first: 2/8, 2/6, 2/4, 2/3. The second fraction is 2/6.
p4c_s1	Set 2: Find the mistake. Ruthie drank 1/2 of a small cup of lemonade. Calvin drank 1/4 of a big jug. Ruthie says she drank more because 1/2 > 1/4. What went wrong? [The wholes are different sizes, so you cannot tell just by comparing the fractions. / Nothing is wrong. 1/2 is always more. / 1/4 is greater than 1/2. / She should compare the numerators, 1 and 1, so they drank the same.]	**The wholes are different sizes, so you cannot tell just by comparing the fractions.**	1/2 > 1/4 only when the wholes are the same size. A small cup and a big jug are different wholes.
p4a_s2	Set 3: Two steps! Aunt Loretta has two same-size spools of thread. She used 3/4 of the red spool and 1/4 of the blue spool. Which spool has less thread left? Type the color.	**red** (also accepted: the red spool, red spool, the red one)	Step 1: The red spool has 1/4 left, and the blue spool has 3/4 left. Step 2: 1/4 < 3/4, so the red spool has less left.
p4b_s2	Set 3: The denominator is 2, 3, 4, 6, or 8. Type a fraction with a numerator of 3 that is less than 3/6.	**3/8** (also accepted: 3 / 8, 3 eighths, three eighths, three-eighths)	With 3 pieces each, a fraction is smaller when the pieces are smaller. Only eighths are smaller than sixths, so the answer is 3/8.
p4c_s2	Set 3: Which comparison is true? [1/6 < 1/4 / 5/8 < 3/8 / 2/3 < 2/4 / 4/6 > 4/4]	**1/6 < 1/4**	Sixths are smaller pieces than fourths, so 1/6 < 1/4. In the others, the symbol points the wrong way.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match The greatest of 2/8, 7/8, and 4/8	**7/8**	2 points if matched on the first try.
match_s0_2	Set 1: Match The least of 1/2, 1/3, and 1/6	**1/6**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s0_3	Set 1: Match Equal to 1/2	**2/4**	2 points if matched on the first try.
match_s0_4	Set 1: Match Equal to 1 whole	**6/6**	2 points if matched on the first try.
match_s0_5	Set 1: Match The greater of 3/4 and 3/8	**3/4**	2 points if matched on the first try.
match_s1_1	Set 2: Match numerator	**the top number: how many parts we count**	2 points if matched on the first try.
match_s1_2	Set 2: Match denominator	**the bottom number: how many equal parts make the whole**	2 points if matched on the first try.
match_s1_3	Set 2: Match >	**is greater than**	2 points if matched on the first try.
match_s1_4	Set 2: Match <	**is less than**	2 points if matched on the first try.
match_s1_5	Set 2: Match =	**is equal to**	2 points if matched on the first try.
match_s2_1	Set 3: Match Same denominator	**compare the numerators**	2 points if matched on the first try.
match_s2_2	Set 3: Match Same numerator	**compare the piece sizes**	2 points if matched on the first try.
match_s2_3	Set 3: Match More equal parts in the whole	**smaller pieces**	2 points if matched on the first try.
match_s2_4	Set 3: Match Fewer equal parts in the whole	**bigger pieces**	2 points if matched on the first try.
match_s2_5	Set 3: Match Wholes of different sizes	**cannot compare fairly**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: aunt loretta lives in gee's bend alabama	**Aunt Loretta lives in Gee's Bend, Alabama.**	Aunt Loretta is a name, and Gee's Bend and Alabama are place names, so they get capital letters. Put a comma between the town and the state, and end with a period.
fs2_s0	Set 1: Fix the sentence: we used red blue and yellow fabric	**We used red, blue, and yellow fabric.** (also accepted: We used red, blue and yellow fabric.)	Start with a capital W, put commas between the colors in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: is three eighths less than five eighths	**Is three eighths less than five eighths?**	Start with a capital I. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: tasha calvin and ruthie sewed the quilt	**Tasha, Calvin, and Ruthie sewed the quilt.** (also accepted: Tasha, Calvin and Ruthie sewed the quilt.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: the quilt show is in july	**The quilt show is in July.**	Start with a capital T. July is the name of a month, so it gets a capital J. End the telling sentence with a period.
fs2_s2	Set 3: Fix the sentence: the ferry crosses the alabama river	**The ferry crosses the Alabama River.**	Start with a capital T. Alabama River is the name of a river, so both words get capital letters. End with a period.

BOSS BATTLE: THE BIG QUILT SHOW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Aunt Loretta has two same-size quilt squares. Square A is cut into 4 equal parts, and 3 parts are red. Square B is cut into 8 equal parts, and 3 parts are red. Which square has more red? Type A or B.	**A** (also accepted: square a)	Step 1: Square A is 3/4 red and Square B is 3/8 red. Step 2: Both are 3 pieces, but fourths are bigger than eighths, so 3/4 > 3/8. Square A has more red.
bb2_s0	Set 1: Which explanation is correct? Ruthie says 5/8 > 5/6 because 8 is bigger than 6. [Ruthie is wrong. Both have 5 pieces, but sixths are bigger, so 5/8 < 5/6. / Ruthie is right. A bigger denominator means a bigger fraction. / Ruthie is wrong, because they are equal. / Ruthie is right, because eighths are bigger than sixths.]	**Ruthie is wrong. Both have 5 pieces, but sixths are bigger, so 5/8 < 5/6.**	Same numerator means compare piece sizes. A whole cut into 8 parts has smaller pieces than one cut into 6 parts, so 5/8 < 5/6.
bb1_s1	Set 2: Two steps! Ruthie and Calvin have same-size strips. Ruthie's strip has 8 equal parts, and she sewed 5 of them. Calvin's strip has 4 equal parts, and he sewed 3 of them. Use what you learned about equivalent fractions last week. Who sewed more? Type the name.	**Calvin** (also accepted: calvin sewed more)	Step 1: Change 3/4 into eighths. Each fourth is 2 eighths, so 3/4 = 6/8. Step 2: Compare 6/8 and 5/8. 6/8 > 5/8, so Calvin sewed more.
bb2_s1	Set 2: Find the mistake. Tasha says this list is in order from least to greatest: 1/2, 1/4, 1/8. What went wrong? [She put them from greatest to least. Least to greatest is 1/8, 1/4, 1/2. / Nothing is wrong. The order is correct. / Least to greatest is 1/4, 1/2, 1/8. / The fractions are all equal, so any order works.]	**She put them from greatest to least. Least to greatest is 1/8, 1/4, 1/2.**	All are 1 piece. Eighths are the smallest pieces and halves are the biggest, so least to greatest is 1/8, 1/4, 1/2.
bb1_s2	Set 3: Two steps! Aunt Loretta is sewing a same-size blue square and gold square. The blue square has 1/4 left to sew. The gold square has 3/4 left to sew. Which square has more finished? Type blue or gold.	**blue** (also accepted: the blue square, blue square, the blue one)	Step 1: The blue square is 3/4 finished, and the gold square is 1/4 finished. Step 2: 3/4 > 1/4, so the blue square has more finished.
bb2_s2	Set 3: Which explanation is correct? Why can you compare 2/6 and 5/6 just by looking at the numerators? [The denominators match, so the pieces are the same size. / Because the numerators are both even numbers. / Because sixths are always small. / You cannot compare them at all.]	**The denominators match, so the pieces are the same size.**	Same denominator means same-size pieces, so the fraction with more pieces is greater: 5/6 > 2/6.

BOSS BATTLE: HANGING THE QUILT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Type a fraction in eighths that is greater than 5/8 and less than 7/8.	**6/8** (also accepted: 6 / 8, 6 eighths, six eighths, six-eighths, 3/4)	In eighths, the only numerator between 5 and 7 is 6. So the fraction is 6/8.
bb4_s0	Set 1: Find the mistake. Calvin wrote 2/3 = 2/6 because both fractions have 2 pieces. What went wrong? [Thirds are bigger pieces than sixths, so 2/3 > 2/6. / Nothing is wrong. They are equal. / 2/6 is greater because 6 is bigger. / He should compare the numerators, so 2/3 < 2/6.]	**Thirds are bigger pieces than sixths, so 2/3 > 2/6.**	The same number of pieces is not the same amount when the pieces are different sizes. 2/3 > 2/6.

ID	Question	Answer	Why (explain to student)
bb3_s1	Set 2: Think about last week! Type <, >, or = to compare the fractions. 4/6 ___ 2/3	**=** (also accepted: is equal to, equal to, equals, equal)	Each third is 2 sixths, so 2/3 = 4/6. They land on the same point on the number line, so 4/6 = 2/3.
bb4_s1	Set 2: Four quilters each sewed part of a same-size square: Tasha sewed 3/8, Calvin sewed 3/4, Ruthie sewed 3/6, and Aunt Loretta sewed 3/3. Who sewed the least? [Tasha / Calvin / Ruthie / Aunt Loretta]	**Tasha**	All sewed 3 pieces. The smallest pieces are eighths, so Tasha sewed the least, 3/8.
bb3_s2	Set 3: Write this comparison using fractions and a symbol: four sixths is greater than four eighths.	**4/6 > 4/8** (also accepted: 4/6>4/8, 4/8 < 4/6, 4/8<4/6, 4/6> 4/8, 4/6 >4/8, 4/8< 4/6, 4/8 <4/6)	Four sixths is 4/6, and four eighths is 4/8. Is greater than is >. So 4/6 > 4/8.
bb4_s2	Set 3: Find the mistake. Tasha says 3/4 < 2/4 because a bigger numerator makes the pieces smaller. What went wrong? [The same denominator means same-size pieces. 3 pieces is more than 2, so 3/4 > 2/4. / Nothing is wrong. / They are equal because both are fourths. / 2/4 is greater because 2 is less than 4.]	**The same denominator means same-size pieces. 3 pieces is more than 2, so 3/4 > 2/4.**	The numerator counts pieces. It does not change the size of the pieces. 3/4 > 2/4.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W18.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 1/4 of a mile in eighths	2/8 of a mile.
2	Calvin sewed 2 of 6 parts; Tasha sewed 5 of 6	Tasha (5/6 > 2/6).
3	1/2 or 1/8 of a same-size square	1/2 is bigger. Fewer equal parts means bigger pieces.
4	Symbol for “is greater than”	>
5	Compare 3/6 and 1/2	3/6 = 1/2. Half of 6 parts is 3 parts, so they land on the same point.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Write <, >, or =	(a) 4/8 < 7/8 (b) 2/3 > 2/4 (c) 1/6 < 1/2 (d) 6/6 = 3/3
2	Tasha used 2/8; Calvin used 5/8. Who has more left?	Tasha has 6/8 left and Calvin has 3/8 left. 6/8 > 3/8, so Tasha has more ribbon left.
3	Calvin wrote 4/6 < 4/8 because 6 is less than 8	With the same numerator, sixths are bigger pieces than eighths. The correct comparison is 4/6 > 4/8.
4	3/4 of a small pillow vs. 1/4 of a big quilt	The wholes are different sizes, so the fractions cannot be compared fairly. 1/4 of a big quilt could be more fabric than 3/4 of a small pillow.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	2/3 vs. 3/6 of same-size strips	Tasha sewed more. 2/3 = 4/6, and 4/6 > 3/6. Drawings should show same-size strips.